CURRICULUM VITAE
Sael Lee

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206 Woldeukeom-ro, Woncheon-dong, Yeongtong-gu,
Suwon-si, Gyeonggi-do, Republic of Korea 08826

Pen name: **Lee Sael**
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RESEARCH INTERESTS

Machine Learning for **Healthcare**, **Tensor Analysis/Data Mining**, and **Bioinformatics**

EDUCATION HISTORY

Purdue University, West Lafayette, IN — Aug. 2005 – Aug. 2010
Ph. D., Computer Science
Dissertation: High throughput screening of global and local protein surfaces

Korea University, Seoul, Republic of Korea — Mar. 2000 – Feb. 2005
Bachelor of Science, Computer Science

AWARDS & FELLOWSHIPS

Best Paper Award (runner up) 6th International Conference on Emerging Databases (EDB) — Oct. 17, 2016
Best Poster Award International Conference on Structure Genomics 2011 — May 14, 2011
Best Paper Award The 21th International Conference on Genome Information (GIW) — Dec. 18, 2010
Grace Hopper Scholarship sponsored by the National Science Foundation — 2008

RESEARCH EXPERIENCE

Professor — Sept. 2019 – Present
Dept. of Software and Computer Engineering, and Dept. of Artificial Intelligence,
Ajou University, Republic of Korea

Visiting Professor (Adjunct Professor) — Aug. 2025 – May 2026
Dept. of Biomedical Engineering, University of South Florida (Tampa), FL, USA

BK Associate Professor — Mar. 2018 – Aug. 2019
Dept. of Computer Science and Engineering, Seoul National University

Assistant Professor — Sept. 2012 – Feb. 2018
Dept. of Computer Science, **State University of New York (Stony Brook) Korea**

Research Staff Member — Sept. 2011 – Aug. 2012
Future IT Research Center, **Samsung Advanced Institute of Technology**

Postdoctoral Research Associate — Sept. 2010 – Aug. 2011
Dept. of Biology, Bioinformatics Laboratory (P.I. Dr. Daisuke. Kihara)
Purdue University, West Lafayette, IN
Funding sources: National Science Foundation (DMS800568, IIS0915801, EF0850009)

Research Assistant — Aug. 2006 – Aug. 2010
Dept. of Computer Science, Bioinformatics Laboratory (P.I. Dr. Daisuke. Kihara)
Purdue University, West Lafayette, IN
Funding source: National Institute of General Medical Sciences of the NIH (R01 GM075004)

SELECTED CURRENT/PAST EXTERNAL RESEARCH GRANTS (PI/CO-PI)

[12] Co-PI: [2022.04-**2026.12**] ***Human-centered Artificial Intelligence Core Technology Development Project*** (Part 4) (사람중심인공지능핵심원천기술개발사업) Development of AI Technology to support Expert Decision-making that can Explain the Reasons/Grounds for Judgment Results based on Expert Knowledge, supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korean government (MSIT) [RS-2022-II220369]

[11] PI: [2022.06-2025.02] ***Basic Science Research Program***, "Uncertainty- and Structure-aware Graph Learning Methods for Supporting Trust-Improved Computational Drug Development," supported by the National Research Foundation of Korea Grant funded by the Korean government [2022R1F1A1065664].

[10] Co-PI: [2018.08-2025.02] ***Engineering Research Center (ERC),*** "Extreme Exploitation of Dark Data (NRF-2018R1A5A1060031)," supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT).

[9] PI: [2018.09-2021.08] ***Basic Science Research Program***, "Interpretable and Dynamic Tensor Decomposition Method Development and Applications to Personalized Clinical Analysis (NRF-2018R1A1A3A0407953)," supported by the National Research Foundation of Korea (NRF) funded by the Ministry of Science, ICT and Future Planning.

[8] PI: [2019.09-2020.12] *Technology Advancement Research Program (TARP),* "Deep Convolution Neural Network based Automated Performance Diagnosis System for Facilities (19CTAP-C152020-01)," funded by Ministry of Land, Infrastructure and Transport of Korean government.

[7] PI: [2019.09-2020.12] *Technology Advancement Research Program (TARP),* "Development of Deep learning-Based Fatigue-Risk Integrated Manpower Monitoring Technology to Prevent Construction Workers Accidents (19CTAP-C152017-01)," funded by Ministry of Land, Infrastructure and Transport of Korean government.

[6] PI: [2015.11-2018.10] *Basic Science Research Program*, "Development of Multi-layer Network Analysis Methods for Integrative Analysis Algorithms of Bio-Clinical Data (NRF-2015R1C1A2A01055739)," supported by the National Research Foundation of Korea (NRF) funded by the Ministry of Science, ICT and Future Planning.

[5] PI: [2015.06- 2018.02] *Global ATC Program* "Development of business intelligence platform service which enables a non-expert to realize 'Data Processing and User Interface' within 3 seconds in Petabyte level based on 0.5 billion data (10053204)," supported by the Korea Evaluation Institute of Industrial Technology, Republic of Korea.

[4] Co-PI: [2015.03-2015.08] "Development of biomedical data network analysis technology based on high performance computing for dementia researches (K-15-L03-C02-S01)," supported by the Korea Institute of Science and Technology Information.

[3] PI: [2012.06-2015.05] *Basic Science Research Program,* supported by the National Research Foundation of Korea (NRF) funded by the Ministry of Science, ICT and Future Planning (2013005259).

[2] Co-PI: [2012.09-2018.02] *IT Consilience Creative Program*, supported by the MSIP (Ministry of Science, ICT and Future Planning), Korea, supervised by the NIPA (National IT Industry Promotion Agency) (NIPA-2013-H0203-13- 1001).

[1] Co-PI: [2012.09-2014.08] *Global Research Network Program,* "Development of RAVAT: the tool that uncovers the roles of rare variants and their functions on common diseases with next generation sequencing data (20120910)," supported by the Ministry of Education and Science Technology, Republic of Korea.

PUBLICATIONS

PEER REVIEWED JOURNAL PUBLICATIONS (using name Lee Sael)

[44] Yoon, J., **Sael, L.** (2025). Document Re-ranking with Evidential Neural Networks. IEEE Access. , vol. 13, pp. 161964-161972.

[43] Lee, S.S., Moon, H.W., **Sael, L.** (2024). Block Incremental Dense Tucker Decomposition with Application to Spatial and Temporal Analysis of Air Quality Data. In: **CMES**-Computer Modeling in Engineering & Sciences 139.1.

[42] Ham, K.P. and Sael, L. (2023). Evidential meta-model for molecular property prediction. In: **Bioinformatics** 39 (10), btad604

[41] Jeon, H., .., **Sael, L.** et al. (2023). "AIVariant: a deep learning-based somatic variant detector for highly contaminated tumor samples". In: Experimental & Molecular Medicine 55.8, pp. 1734–1742.

[40] Jang, J.G. Park, M. Lee, J. **Sael, L.** (2022). "Large-scale Tucker tensor factorization for sparse and accurate decomposition". **The Journal of Supercomputing**, pp. 1–31.

[39] Lee, J.W, Choi, M., **Sael, L.,** Shim, H., Lee, J. (2022). Knowledge distillation meets recommendation: collaborative distillation for top-N recommendation. Knowledge and Information Systems 64.5, pp. 1323–1348.

[38] Cho, Eunho .., **Sael, L.** et al. (2022). A neural network model for free-falling condensation heat transfer in the presence of non-condensable gases. International Journal of Thermal Sciences 171, p. 107202.

[37] Shin, H.K. .., **Sael, L.** et al. (2021). Defect-detection model for underground parking lots using image object-detection method. Computers, Materials & Continua 66.3, pp. 2493–2507.

[35] Park, J.Y., .., **Sael, L.** et al. (2021). A missense variant in SHARPIN mediates Alzheimer's disease-specific brain damages. Translational psychiatry 11.1, p. 590.

[34] Lee, H., .., **Sael, L.** et al. (2021). An artificial neural network model for predicting frictional pressure drop in micro-pin fin heat sink. Applied Thermal Engineering 194, p. 117012.

[33] Jung, J. & **Sael, L.** (2020). Fast and accurate pseudoinverse with sparse matrix reordering and incremental approach. Machine Learning 109: 2333–2347.

[32] Lee, K, Hong, G., **Sael, L.,** Lee, S., & Kim, HY. (2020). MultiDefectNet: Multi-Class Defect Detection of Building Façade Based on Deep Convolutional Neural Network Sustainability 12 (22), 9785.

[31] Choi. YJ. Ohn, JH, **Sael, L**, et al. (2020). Family-based exome sequencing combined with linkage analyses identifies rare susceptibility variants of MUC4 for gastric cancer PloS ONE 15 (7), e0236197.

[30] Choi, D. & **Sael, L.** (2019). SNeCT: Scalable Network Constrained Tucker Decomposition for Multi-Platform Data Profiling. IEEE/ACM Transactions on Computational Biology and Bioinformatics (**TCBB**) 17(5): 1785 - 1796. (SCIE Q1; top 9.35%; IF: 2.428)

[29] Lim, Y., Yu, I., Seo,D., Kang, U & **Sael, L.** (2019). PS-MCL: Parallel Shotgun Coarsened Markov Clustering of Protein Interaction Networks. **BMC Bioinformatics** 20(13):#381. (SCIE Q1; top 22.88%; IF: 2.213)

[28] Oh, S., Park. N., Jang. J., **Sael**, **L.** & Kang, U (2019). "High-Performance Tucker Factorization on Heterogeneous Platforms." IEEE Transactions on Parallel and Distributed Systems (**TPDS**), 30(10): 2237–48. (SCI Q1; top 10.4%; IF=3.402)

[27] Lee, J., Oh, S. & **Sael**, **L.** (2018). GIFT: Guided and Interpretable Factorization for Tensors with an Application to Large-Scale Multi-platform Cancer Analysis. **Bioinformatics**, 34(24), 4151–4158. (SCI Q1; top 2.6%; IF=7.307)

[26] Lee, J., Choi, D., & **Sael, L.** (2018). CTD: Fast, Accurate, and Interpretable Method for Static and Dynamic Tensor Decompositions. PLOS ONE 13(7): e0200579. (SCIE Q1; top 22.7%; IF=2.766)

[25] Dehiya, V., & **Sael, L.** (2018). Impact of structural prior knowledge in mutation prioritization: Towards causal variant finding in rare disease. PLOS ONE 13(9):e0204101. (SCIE Q1; top 22.7%; IF=2.766)

[24] Thomas, J., and **Sael, L.** (2017). Multi-Kernel LS-SVM based integration bio-clinical data analysis and application to ovarian cancer. International Journal of Data Mining and Bioinformatics (IJDMB), 19(2) 150-166. (SCIE Q4; IF=0.624)

[23] Shin, K., **Sael, L.,** & Kang, U. (2017). Fully Scalable Methods for Distributed Tensor Factorization. IEEE Transactions on Knowledge and Data Engineering (TKDE) 29(1), 100–113. (SCI Q1; top 14.0%; IF=3.438)

[22] Thomas, J., Seo, D., & **Sael, L**. (2016). Review on graph clustering and subgraph similarity based analysis of neurological disorders. International Journal of Molecular Sciences (IJMS), 17(6), 862. (SCIE Q2; top 32.2%; IF=3.226)

[21] Jeon, I., Papalexakis, E. E., Faloutsos, C., **Sael, L**, Kang, U. (2016). Mining billion-scale tensors: algorithms and discoveries. The VLDB Journal, 1-26. (SCI Q1; top 5.1%; IF=4.269)

[20] Jung, J., Shin, K., Sael, L., & Kang, U. (2016). Random walk with restart on large graphs using block elimination. ACM Transactions on Database Systems (TODS), 41(2), 1–43. (SCI)

[19] Kim, S., **Sael, L**.*, & Yu, H.* (2015). A mutation profile for top-k patient search exploiting Gene-Ontology and orthogonal non-negative matrix factorization. **Bioinformatics**, 31(22), 3653-3659. (SCI Q1; top 2.6%; IF=7.307)

[18] Kim, S., **Sael, L.,** & Yu, H. (2015). LMDS-based approach for efficient top-k local ligand-binding site search. International Journal of Data Mining and Bioinformatics (IJDMB), 12(4), 417–433. (SCIE Q4; IF= 0.624)

[17] **Sael, L**., Jeon, I., & Kang, U. (2015). Scalable tensor mining. Big Data Research, 2(2), 82–86.

[16] Pi, J., **Sael, L.** (2013) Mass spectrometry coupled experiments and protein structure modeling methods. International Journal of Molecular Sciences, 14:10, 20635-20657. (SCIE Q2; top 32.2%; IF=3.226)

[15] Kim, S., **Sael, L.**, Yu, H. (2013). Efficient protein structure search using indexing methods. BMC Medical Informatics and Decision Making, 13 (Suppl 1), S8. (SCIE Q2; IF=1.496).

[14] Mullins, E. A., Starks, C. M., Francois, J. A., **Sael, L.**, Kihara, D., & Kappock, T. J. (2012). Formyl-coenzyme A (CoA):oxalate CoA-transferase from the acidophile Acetobacter aceti has a distinctive electrostatic surface and inherent acid stability. Protein Science, 21:5, 686-696. (SCI Q3; IF=2.735)

[13] **Sael, L.**, Chitale, M., & Kihara, D. (2012). Structure and sequence-based function prediction for non-homologous proteins. Journal of Structural and Functional Genomics, 13:111-123.

[12] **Sael**, **L.,** & Kihara, D. (2012). Detecting local ligand-binding site similarity in non-homologous proteins by surface patch comparison. **Proteins**: Structure, Function, and Bioinformatics, 80:4, 1177-1195. (SCI Q2; IF=3.337)

[11] **Sael**, **L.,** & Kihara, D. (2012). Constructing patch-based ligand-binding pocket database for predicting function of proteins. **BMC Bioinformatics**, 13(Suppl 2):S7. (SCIE Q1; IF=3.024)

[10] Kihara, D., **Sael**, **L.,** Chikhi, R., & Esquivel-Rodriguez, J. (2011). Molecular surface representation using 3D Zernike descriptors for protein shape comparison and docking. Current Protein and Peptide Science, 12:6, 520-530.(SCIE Q2; IF=2.886)

[9] Chikhi, R., **Sael**, **L.,** & Kihara, D. (2010). Real-time ligand binding pocket database search using local surface descriptors. Proteins: Structure, Function, and Bioinformatics, 78:9, 2007-2028. (SCI Q2; IF=3.337)

[8] **Sael**, **L.,** & Kihara, D. (2010). Binding ligand prediction for proteins using partial matching of local surface patches. International Journal of Molecular Sciences, 11:12, 5009-5026. (SCIE Q2; top 32.2%; IF=3.226)

[7] **Sael**, **L.,** *&* Kihara*, D.* (2010). Improved protein surface comparison and application to low-resolution protein structure data. **BMC Bioinformatics**, 11 (Suppl 11): S2, 1-12. (SCIE Q1; IF=3.024) **GIW2010 - BEST PAPER AWARD**

[6] **Sael**, **L.,** *&* Kihara*, D.* (2010). Characterization and classification of local protein surfaces using self-organizing map. International Journal of Knowledge Discovery in Bioinformatics, 1:1, 32-47

[5] La, D., Esquivel, J., Venkatraman, V., Li, B., **Sael**, **L.,** Ueng, S., Ahrendt, S., & Kihara, D. (2009). 3D-SURFER: software for high-throughput protein surface comparison and analysis. **Bioinformatics***,* 25:21, 2843-2844. (SCI Q1; IF=5.766)

[4] Venkatraman, V., **Sael, L.,** *&* Kihara, D. (2009). Potential for protein surface shape analysis using spherical harmonics and 3D Zernike descriptors. Cell Biochemistry and Biophysics, 54:1-3, 23-32. (SCIE Q2; IF=3.337)

[3] Venkatraman, V., Yang, Y., **Sael, L.,** & Kihara, D. (2009). Protein-protein docking using region-based 3D Zernike descriptors. **BMC Bioinformatics**, 10:407, 1-21. (SCIE Q1; IF=3.024)

[2] **Sael**, **L.,** La, D., Li, B., Rustamov, R., & Kihara, D. (2008). Rapid comparison of properties on protein surface. **Proteins** 73:1, 1-10. (SCI Q2; IF=3.337)

[1] **Sael**, **L.,** Li, B., La, D., Fang, Y., Ramani, K., Rustamov, R., & Kihara, D. (2008). Fast protein tertiary structure retrieval based on global surface shape similarity. **Proteins**, 72:4, 1259-1273. (SCI Q2; IF=3.337)

INTERNATIONAL CONFERENCE PROCEEDINGS (using name Lee Sael)

[20] Yoo, Jaemin and Lee Sael (2022). Transition Matrix Representation of Trees with Transposed Convolutions. In: **SIAM International Conference on Data Mining (SDM22). (BK IF=2)**

[19] Ham, KyungPyo, JeongNoh Yoon, and Lee Sael (2022). Towards accurate and certain molecular properties prediction. In: 2022 13th International Conference on Information and Communication Technology Convergence (ICTC). IEEE, pp. 1621–1624.

[18] Park, Jin-Jun and Lee Sael (2022). Benchmarking Deep Graph Models for Large Molecular Generation. In: 2022 IEEE International Conference on Big Data and Smart Computing (BigComp). IEEE, pp. 114–120.

[17] Heesoo, Son, Lee Sangseok, and Lee Sael (2022). Cross-Attention Model for Multi-modal Bio-Signal Processing. In: 2022 IEEE International Conference on Big Data and Smart Computing (BigComp). IEEE, pp. 43–46.

[16] Yoo, Jaemin and Lee Sael (2021). Gaussian Soft Decision Trees for Interpretable Feature-Based Classification. In: The Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD)**. Vol. 2, pp. 143–155. **(BK IF=1)**

[15] Part, M., Jang, JG & **Sael, L** (2021). VEST: Very Sparse Tucker Factorization of Large-Scale Tensors. In BigComp2021. Jeju, Korea.

[14] Yoo J. & **Sael, L** (2019) EDiT: Interpreting Ensemble Models via Compact Soft Decision Trees. In the IEEE International Conference on Data Mining (**ICDM** 2019). Beijing, China. **(BKCSA061 IF=3)**

[13] Oh, S., Park, N., **Sael, L**., & Kang, U., (2018) Scalable Tucker Factorization for Sparse Tensors - Algorithms and Discoveries. In 34th IEEE International Conference on Data Engineering (**ICDE** 2018). **(BKCS0051 IF=3)**

[12] Thomas, J., Thomas, S., **Sael, L.** (2017) DP-miRNA: An Improved Prediction of precursor microRNA using Deep Learning Model. BIGCOMP 2017. Jeju, Korea.

[11] Jung, Jinhong, Namyong Park, Lee Sael, and U Kang. 2017. BePI: Fast and Memory-Efficient Method for Billion- Scale Random Walk with Restart. In ACM International Conference on Management of Data (**SIGMOD** 2017). Raleigh, North Carolina, USA.: ACM Press. (**BKCSA028 IF=4**)

[10] Thomas, J., & **Sael, L**. (2016). Maximizing information through multiple kernel-based heterogeneous data integration and applications to ovarian cancer. In 6th International Conference on Emerging Databases. **BEST PAPER** (Runner-Up)

[9] Jeon, B., Jeon I., **Sael, L.**, & Kang, U. (2016) SCouT: Scalable coupled matrix-tensor factorization - Algorithm and Discoveries. In 32nd IEEE International Conference on Data Engineering (**ICDE** 2016). **(BKCS0051 IF=3)**

[8] Thomas, J., & **Sael, L.** (2015). Overview of integrative analysis methods for heterogeneous data. In 2015 International Conference on Big Data and Smart Computing (BIGCOMP) (pp. 266–270).

[7] Shin, Kijung, Jinhong Jung, Lee Sael, and U Kang. (2015). "BEAR: Block Elimination Approach for Random Walk with Restart on Large Graphs." In Proceedings of the 2015 ACM International Conference on Management of Data (**SIGMOD** 2015), 1571–85. Melbourne, Victoria, Australia: ACM Press. (**BKCSA028 IF=4**)

[6] Kang, D., Lim, W., Shin, K., **Sael, L**., & Kang, U. (2014). Data/Feature Distributed Stochastic Coordinate Descent for Logistic Regression. In Proceedings of the 23rd International Conference on Information and Knowledge Management (**CIKM** 2014). Shanghai, China: ACM. **(BKCS0027 IF=3)**

[5] Kim, S., **Sael, L**., & Yu, H. (2014). Identifying cancer subtypes based on somatic mutation profile. In Proceedings of the 8th International Workshop on Data and Text Mining in Biomedical Informatics - DTMBIO'14. Shanghai, China: ACM.

[4] Coelho, D., & **Sael, L**. (2013). Breast and prostate cancer expression similarity analysis by iterative SVM based ensemble gene selection. In Proceedings of the 7th International Workshop on Data and Text Mining in Biomedical Informatics - DTMBIO '13 (pp. 23–26). San Francisco, USA: ACM.

[3] Kim, S., **Sael, L.,** & Yu, H. (2013) Efficient local ligand-binding site search using landmark MDS. Proceedings of the 7th International Workshop on Data and Text Mining in Biomedical Informatics – DTMBIO'13. San Francisco, USA: ACM.

[2] Kim, S., **Sael, L.,** & Yu, H. (2013) Fast protein 3D surface search. Proceedings of the 7th International Conference on Ubiquitous Information Management and Communication - ICUIMC'13 (pp. 1–8). Kota Kinabalu, Malaysia: ACM.

[1] Kim, S., **Sael, L.**, & Yu, H. (2012). Indexing methods for efficient protein 3D surface search. In Proceedings of the ACM 6th International Workshop on Data and Text Mining in Biomedical informatics - DTMBIO '12 (p. 41). New York, New York, USA: ACM.

BOOK CHAPTERS (using name Lee Sael)

[4] Xiong, Y., Esquivel-Rodriguez, J., **Sael, L.,** & Kihara, D. (2014). 3D-SURFER 2.0: Web platform for real-time search and characterization of protein surfaces. Methods in Molecular Biology, 1137, 105–117.

[3] Tang, M., Tan, K. M., Tan, X. L., **Sael, L**., Chitale, M., Esquivel-Rodríguez, J., & Kihara, D. (2013). Graphical models for protein function and structure prediction. In M. Elloumi & A. Y. Zomaya (Eds.), Biological Knowledge Discovery Handbook: Preprocessing, Mining, and Postprocessing of Biological Data (pp. 191-222).. Hoboken, New Jersey: John Wiley & Sons, Inc.

[2] Chikhi, R., **Sael, L.,** & Kihara, D. (2011). Protein binding ligand prediction using moment-based methods. In D. Kihara (Ed.), Protein Function Prediction for Omics Era (pp. 145–163). Dordrecht: Springer Netherlands.

[1] **Sael, L.**, & Kihara, D. (2009). Protein surface representation and comparison: New approaches in structural proteomics. In J. Y. Chen & S. Lonardi (Eds.), Biological Data Mining (pp. 89–109). USA: Chapman & Hall/CRC Press

CONFERENCE PRESENTATIONS (**selected** posters or orals at conferences that do not have proceedings)

[6] (*poster*) **Sael Lee**, J. Thomas, Y. Shin, H. Yoon, Y. Park. "Data Mining Bio-clinical Data." at World IT Show, May 17-20, 2016, Seoul, Korea.

[5] (*poster*) **Sael, L.,** & Kihara, D. "Binding ligand prediction by comparing local surface patches of potential pocket regions." at International Conference on Structural Genomics, May 10-14, 2011, Toronto, Canada. ***BEST POSTER***

[4] *(poster)* **Sael, L.,** & Kihara, D. (2011). Local Protein Surface Patch Method for Protein-Ligand Binding Prediction. BIOPHYSICAL JOURNAL 100, 3, 1: 394. (Mar. 5-6, 2011, 55th Annual Meeting of the Biophysical-Society Location: Baltimore, MD)

[3] *(poster)* **Sael. L.**, Esquivel-Rodriguez, J., & Kihara, D. (2011). Novel Methods for Rapid Comparison and Multimeric Protein Complex Fitting for Low-Resolution Electron Microscopy Data. BIOPHYSICAL JOURNAL 100, 3, 1: 323. (Mar. 5-6, 2011, 55th Annual Meeting of the Biophysical-Society Location: Baltimore, MD)

[2] *(oral)* **Sael, L.,** & Kihara, D."Patch-Surfer: Alignment free surface patch-based ligand binding pocket comparison." at Great Lakes Bioinformatics Conference 2011, May 2-4, 2011, Ohio University, Athens, Ohio

[1] *(oral)* **Sael, L.,** & Kihara, D."Improved protein surface comparison and application to low-resolution protein structure data." The 21st International Conference on Genome Informatics, Hangzhou, China, December 16-18, 2010

TALKS

TUTORIALS

[7] Sael Lee. "Interpretable Machine Learning in Biomedical Data Analysis." Bioinformatics & Machine Learning Workshop for Life Scientists (BIML), Seoul National University, Korea, Jan.31, 2020.

[6] Sael Lee, "An Introduction to **Interpretable Machine Learning**." *Korea Software Congress (KSC)* 2018, PyeongChang, Korea. Dec. 12-19, 2018.

[5] Sael Lee, "Tutorial on AI/ML/DL." *The Korean Institute of Broadcast and Media Engineers (KIBME) Spring Workshop* May 31, 2018. Seoul Korea.

[4] Sael Lee, "AI/Machine Learning/**Deep Learning** Tutorial and Current Research Trends." *Open Standards and Internet Associations* (*OSIA) Workshop* Nov. 30, 2017. Seoul, Korea .

[3] Sael Lee, "**Interpretable Deep Learning** and Applications to Bio-healthcare." *Open Standards and Internet Associations* (*OSIA) Workshop* June 30, 2017. Konkuk University.

[2] Sael Lee. "**Deep Learning in BioHealth**: Detailing **Interpretability**." *Korea Computer Congress (KCC)* 2017, Jeju Island, Korea. June 18-20, 2017.

[1] Sael Lee. "Mining and Learning **Bio-Big Data**." *Bioinformatics & Machine Learning for Life Scientists Workshop*, Seoul National University, Korea, Feb. 16, 2017.

INVITED CONFERENCE/WORKSHOP SPEAKER

[12] Jian Pei, Stephan Pareigis, Tok Wang Ling, Sael Lee, & Ho-Jin Choi, "Explainability in Big Data Analysis: essential, redundant or nice to have." BigComp2021 Panel Talk#2, Jeju Island, Korea. Jan. 19, 2021.

[11] Sael Lee. "Interpretable Multi-dimensional Data Analysis with Tensor Mining." AI Korea 2019, Yonsei University, Seoul, Korea, July 25-26, 2019.

[10] Sael Lee. "Deep Learning in Computational Biology." BIOINFO2019, Ewha Womans University, Seoul, Korea, Aug. 26 - 27, 2019.

[9] Sael Lee, "Scalable Network Constrained Tensor Mining." Korea AI Society Workshop jointly held with ACML 2017. Seoul, Korea.

[8] Sael Lee, "Scalable Tensor Factorization and Applications in Bioinformatics." GIW/BIOINFO 2017. Seoul, Korea.

[7] Sael Lee, "Deep learning applications published in my favorite journals." Annual Korean Society for Bioinformatics Conference, Aug. 18, 2016. Incheon Global Campus.

[6] Sael Lee, "Orthogonal NMF-based Top-K Patient Mutation Profile Searching." Current Trends in AI Workshop, June 29, 2016. Jeju, Korea.

[5] Sael Lee, "Using functional information to prioritize single nucleotide polymorphisms." Conference on BioMedical Informatics. Nov. 12, 2013. Busan, Korea.

[4] Sael Lee, "Towards Computational Screening of Global and Local Protein Surface for Drug Discovery." jCB Symposium on Frontier in Structural Biology of Cells and Molecules. Dec. 7, 2012. Incheon, Korea.

[3] Sael Lee, "AI in the Clinics: Clinical Decision Support System." Annual Conference of BioMedical Society. Nov. 23, 2012. Ohsong, Korea.

[2] Sael Lee, "Drug development in the perspective of structural bioinformatics." Korea Information Processing Society Biomedical and Information Technology Workshop, Dec. 13, 2011. Korea University, Seoul, Korea

[1] Sael Lee, "Structural Bioinformatics." NewDB Workshop, Sept. 24, 2011, National Information Society Agency (NIA).

TEACHING EXPERIENCE

Undergraduate/ Graduate Core Courses

CSE 313 Data Mining	**Ajou University**
CSE 231 Algorithm	**Ajou University**
CSE 333 Database	Ajou University
IS12211 Big Data (Python Programming)	Ajou University
DS 511 Data Statistics	Ajou University
AI 501 Human Centered Artificial Intelligence	Ajou University
DS 512 Computer Programing in Artificial Intelligence	Ajou University
DHC 519 Medical Humanities and Text-Mining	Ajou University
DS 722 Advanced Topics in Machine Learning I	Ajou University
DS 723 Advanced Topics in Machine Learning II	Ajou University
BME 4931/6931 BME Machine Learning	University of South Florida ('25-'26)
CSE 214 Computer Science II (Data Structure)	SUNY Korea
CSE 305 Principles of Database Systems	SUNY Korea
CSE 373 Analysis of Algorithms	SUNY Korea
CSE 532 Theory of Database Systems	SUNY Korea
CSE 537 Artificial Intelligence	SUNY Korea
CSE 548 Analysis of Algorithms	SUNY Korea
CSE 549 Computational Biology	SUNY Korea

Project/Seminar Courses

CSE 600 Topics in Modern Computer Science	SUNY Korea
CSE 646 Seminar on Artificial Intelligence	SUNY Korea
CSE 523 Advanced Project in Computer Science I	SUNY Korea
CSE 524 Advanced Project in Computer Science II	SUNY Korea

Special Courses

Basic Math for Big Data (Discrete Math)	SNU 4th Industrial Revolution Academy (SNU FIRA)

PROFESSIONAL SERVICES

Member of board of directors:

KIISE Artificial Intelligence Society	2013 – present
Korean Society for Bioinformatics (KSBi)	2015 – present
KIISE Database Society of Korea	2016 – present

Guest editor:

Journal of Bioinformatics and Computational Biology (JBCB)	2017

Local chair:

Annual Korean Society for Bioinformatics Conference (BIOINFO2016)	2016

Organizing committee:

2017 Korean AI Research Council Workshop **"**AI for Everyone"	2017

Program committee:

World Wide Web Conference (**WWW**)	2018
Pacific-Asia Conference on Knowledge Discovery and Data Mining **(PAKDD)**	2018 – 2019
ACM International Conference on Information and Knowledge Management (**CIKM**)	2017
International Conference on Genome Informatics (**GIW**) / BIOINFO	2017
Translational Bioinformatics Conference (**TBC**)	2015 – 2017
International Conference on Emerging Databases (EDB)	2016
Workshop on Current Trends in AI	2016
WWW 2014 Workshop on Big Graph Mining (BGM 2014)	2014
ACM International Workshop on Data and Text Mining in Biomedical Informatics (DTMBIO)	2012 – 2018

Session chair for many conferences and workshops 2012 – present

Reviewer for many journal and conferences, including the following:
CIKM, RECOMB, BIBM, IJMS, TBC, and TCBB, Bioinformatics, BMC Bioinformatics,.

ACADEMIC MEMBERSHIP

Member of Korean Institute of Information Scientists and Engineers (KIISE)	2012 – present
Member of International Society for Computational Biology (ISCB)	2011 – present
Member of Biophysical Society	2011 – 2015
Member of The Protein Society	2008 – 2015
Member of Association for Computing Machinery (ACM)	2004 – present